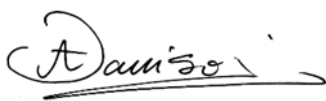


Approved By Associate Dean:



Jan 2, 2025

Signature

COURSE SECTION INFORMATION

Full Stack Data Science Systems Applied A.I. Solutions Development

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

The focus of this course will be on the development of full stack data science projects. Students will work on a showcase project reflecting the culmination of skills acquired throughout the year. Topics will include the integration of model building, data visualisation, data management, model tuning, and model optimisation within a seamless, end-to-end AI project.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Develop and manage a seamless end-to-end AI system.
2. Communicate insights and recommendations to stakeholders through the use of comprehensive, professional dashboards.
3. Integrate the knowledge acquired from previous modules such as model building, model tuning, model optimisation into a working project
4. Deploy a fully trained and optimised model in a cloud environment.
5. Connect visualisation tools to deployed models.
6. Connect an API to a deployed model to offer it as a data service.

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Required:**Recommended Resources:****Important Note on the Use of Generative AI:**

Students must review the "Generative AI Usage Guidelines" document, available on D2L, or consult with the instructor for details on how generative AI tools may be used in this course.

Generally, use of AI is allowed and encouraged in most of the courses of this program, this include projects and assignments. However use of AI are not allowed in exams and quizzes unless specified otherwise by the instructor.

Students must consult their instructor when unsure.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Participation	Attendance and in-class participation in various activities	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1 - 3	10
Lab Exercises 10 @ 3%	Hands-on exercises	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1 - 3	30
Project	Individual Assignments	1,2,3,4,5,6,7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	3	60
TOTAL:					100%

GRADING SYSTEM the passing grade for this course is: D (50%)

A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Excerpt from the College Policy on Academic Dishonesty:

The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.

To view George Brown College policies please go to www.georgebrown.ca/policies

Learning Schedule / Topical Outline (subject to change with notification)

TOPICAL OUTLINE:

Week	Topic Task	In-person/Online	Outcome(s)	Content / Activities	Resources
1	1	Monday In-person	1, 4, 5	ML Canvas	Resource material available on D2L
1	2	Wednesday Online	1, 4, 5	Scripting Notebooks	Resource material available on D2L
1	3	Friday Online	1, 4, 5	Flask	Resource material available on D2L
Tasks: Lab Exercises 1, 2, 3, 4					
2	4	Monday In-person	1, 2, 4, 5, 6	Clients	Resource material available on D2L
2	5	Wednesday In-person	1, 2, 4, 5, 6	Docker	Resource material available on D2L
2	6	Friday Online	1, 2, 4, 5, 6	Cloud Deployment	Resource material available on D2L
Tasks: Lab Exercises 5, 6, 7, 8, 9, 10					
3	7	Monday In-person	1, 2, 3, 4, 5, 6	Case Studies - I	Resource material available on D2L

3	8	Wednesday Online	1, 2, 3, 4, 5, 6	Case Studies - II	Resource material available on D2L
3	9	Friday In-person	1, 2, 3, 4, 5, 6	Project Demo Presentation	Resource material available on D2L
Tasks: Project Due					
Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx					